ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice e.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 A ✓ 1.6 B ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 True ✓ 2.3 False ✓ 2.4 True ✓ 2.5 False ✓ 2.6 True ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 False ✓ (10)

[10]

QUESTION 3

3 3.1 F ✓ 3.2 C ✓ 3.3 H ✓ 3.4 D ✓ 3.5 A ✓ 3.6 G ✓ 3.7 I ✓ 3.8 B ✓ 3.9 J ✓ 3.10 E ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Hold the thumb, first finger and second finger of the left hand at right angles to each other ✓: the First finger points in the direction of the magnetic Field ✓, the seCond finger in the direction of the Current, and the thuMb gives the direction of the Motion (force) ✓. (3)

4.2 $F = B \times I \times l$ ✓ $= 0{,}5 \times 6 \times 0{,}25$ ✓ $= 0{,}75$ N ✓ (3)

4.3 Any FOUR ✓✓✓✓: increase the current; increase the number of turns; use a soft-iron core; make the core larger / shorter gap; wind the turns closer together. (4)

4.4 Current through the coil makes it an electromagnet ✓; it attracts the armature ✓, which moves the contacts to open or close the separate switched circuit ✓. (3)

4.5 Any TWO ✓✓: relays and contactors; solenoid valves; electromagnetic grippers for steel parts; electric locks; motors; electromagnetic brakes. (2)

[15]

QUESTION 5

5.1 $V_2 = V_s \times R_2 \div (R_1 + R_2)$ ✓ $= 10 \times 3 \div 5$ ✓ $= 6\ V$ ✓ (3)

5.2 $I = 10 \div 5\,000$ ✓ $= 0{,}002\ A\ (2\ mA)$ ✓ (2)

5.3 $\eta = 400 \div 500 \times 100$ ✓ $= 80\%$ ✓; $losses = 500 - 400 = 100\ W$ ✓ (3)

5.4 $E = 2 \times 3$ ✓ $= 6\ kWh$ ✓; $cost = 6 \times 2{,}80$ ✓ $= R16{,}80$ ✓ (4)

5.5 When the armature turns in the magnetic field, an emf is induced in it that opposes the supply voltage – the back emf ✓✓. At start-up the armature is not turning, so there is no back emf and only the small armature resistance limits the current ✓, which makes it very large ✓. (4)

5.6 $I = 12 \div (4 + 8)$ ✓ $= 1\ A$ ✓; $V_1 = 1 \times 4 = 4\ V$ ✓; $V_2 = 1 \times 8 = 8\ V$ ✓ (4)

[20]

QUESTION 6

6.1 N-type: pure silicon is doped with a pentavalent impurity (5 valence electrons, e.g. phosphorus or arsenic) ✓, giving extra free electrons ✓. P-type: silicon is doped with a trivalent impurity (3 valence electrons, e.g. boron or gallium) ✓, giving holes ✓. (4)

6.2 An LED gives off light when current flows through it in the forward direction ✓ – e.g. status / indicator light, IR transmitter on a line sensor ✓. A photodiode is used in reverse bias and lets more current flow when light falls on it ✓ – e.g. light sensor, IR receiver, encoder ✓. (4)

6.3 Zener diode symbol (bar with bent ends) ✓; photodiode symbol with arrows pointing INTO it ✓; PNP symbol with the emitter arrow pointing INWARDS ✓ (3)

6.4 Without a resistor the current would be too large and the LED would burn out ✓; $R = (9 - 2) \div 0{,}02$ ✓✓ $= 350\ \Omega$ ✓ (4)

[15]

QUESTION 7

7.1.1 $45 = 32 + 8 + 4 + 1$ ✓ $= 101101_2$ ✓ (2)

7.1.2 $16 + 8 + 0 + 2 + 1$ ✓ $= 27$ ✓ (2)

7.1.3 $2 \times 16 + 15$ ✓ $= 47$ ✓ (2)

7.2 Correct NOR symbol (OR shape with a circle at the output) ✓; truth table ✓✓✓✓: (5)

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

7.3 Q = A(1 + B) ✓ = A · 1 = A ✓; absorption law ✓ (3)

7.4 Any FOUR in order ✓✓✓✓: switch off and disconnect the power; heat the joint and remove the solder with a desoldering pump or braid; remove the faulty component, noting its orientation; clean the pads; fit the new component the correct way round; solder the joints neatly and cut off extra leads; inspect and test the circuit. (4)

7.5 Any TWO ✓✓: joint not heated enough (iron too cold or removed too soon); dirty or oxidised pad / lead; the joint moved while cooling; too little solder or flux. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE